BlockChain Sample Ques

Q1: Apply the CAP theorem to analyze the trade-offs in a distributed blockchain network during a network partition.

CAP theorem and blockchain during a network partition

1. CAP theorem
A distributed system can guarantee only two of these three properties at the same time:

- **Consistency (C):** every node sees the same data (the same ledger).
- **Availability (A):** every request gets a response, so the network keeps accepting transactions.
- **Partition tolerance (P):** the system keeps working even when messages between nodes are lost or delayed.

A network partition can always happen on the internet, so P cannot be given up. When a partition occurs, the real choice is between **C and A**.

2. What happens in a blockchain during a partition
Suppose the network splits into two groups, A and B, that cannot talk to each other.

- Each group keeps accepting transactions and mining its own blocks.
- Each group builds its own version of the ledger, so the **chain forks**.
- Group A may record "Alice pays Bob" while group B records "Alice pays Carol" (a double spend).

3. Where blockchains sit in CAP

Type	Choice	Behaviour during a partition
Bitcoin / Ethereum (public, PoW)	**AP**	Both sides keep working, so availability is kept, but the ledgers differ for a while. Consistency is only *eventual*.

Type	Choice	Behaviour during a partition
Hyperledger Fabric / permissioned (PBFT, Raft)	**CP**	The side without a quorum (majority) stops committing blocks. This keeps the ledger consistent but sacrifices availability.

4. How consistency is restored after the partition heals

- Public chains use the **longest-chain (most work) rule**. The shorter fork is discarded and its transactions go back to the mempool. Waiting for several confirmations before treating a payment as final is how this risk is handled.
- This is **eventual consistency**: the ledger is not consistent during the partition, but all nodes converge afterwards.
- Permissioned chains avoid forks by requiring a quorum before any block is committed.

5. Trade-offs

	Favour Availability (AP)	Favour Consistency (CP)
Pros	Network never stops; open and decentralized	No forks or double spends; instant finality
Cons	Temporary forks, double-spend risk, rollbacks	Halts if quorum is lost; needs known participants
Best for	Cryptocurrencies	Enterprise / banking use cases

6. Conclusion

Because partitions cannot be avoided, a blockchain must choose between consistency and availability. Public blockchains such as Bitcoin choose **availability with eventual consistency (AP)**. Permissioned blockchains such as Hyperledger Fabric choose **consistency over availability (CP)**. The right choice depends on whether the application values uptime or strict correctness.

Q2: A startup is selling digital land parcels in a virtual estate platform, and each parcel is unique. Which token standard is

most suitable?

Answer: ERC-721 (Non-Fungible Token standard)

Why ERC-721 fits

- **Each parcel is unique.** ERC-721 gives every token its own `tokenId`, so no two parcels are the same and each one can be tracked separately.
- **Clear ownership.** `ownerOf(tokenId)` returns the single owner of a parcel, so ownership is public and verifiable on the blockchain.
- **Safe transfers.** `transferFrom` and `safeTransferFrom` let a parcel be sold or gifted. `approve` lets a marketplace sell it on the owner's behalf.
- **Metadata.** `tokenURI` links each parcel to its details (location, coordinates, size, image), which are usually stored off-chain, e.g. on IPFS.
- **Wallet and marketplace support.** ERC-721 works with MetaMask and NFT marketplaces such as OpenSea.

Why not the others

Standard	Type	Why it doesn't fit
ERC-20	Fungible	All tokens are identical and interchangeable, like currency, so it cannot represent unique parcels.
ERC-1155	Multi-token (fungible and non-fungible)	It could work, but it is better suited to games with many item types. It is more than a simple land registry needs.
ERC-721	Non-fungible	One token is one unique asset, which matches one parcel.

Conclusion: ERC-721 is the most suitable standard because it makes every digital land parcel a unique, individually owned and tradable token.

Q3: Define Hyperledger Fabric and explain its suitability for enterprise blockchain applications.

Hyperledger Fabric

Definition
Hyperledger Fabric is an open-source, **permissioned** and **modular** blockchain platform hosted by the Linux Foundation. It is built for enterprise use: only known, authorized participants can join the network, and business logic runs as **chaincode** (smart contracts, written in Go, Java or JavaScript).

Key components

- **Organizations and peers:** peers hold the ledger and run chaincode.
- **Ordering service:** orders transactions into blocks.
- **Channels:** private sub-networks between selected organizations.
- **Membership Service Provider (MSP):** manages identities and certificates.
- **Ledger:** the world state (current data) plus the transaction history.

Transaction flow
Client proposes → peers endorse → ordering service creates blocks → peers validate and commit.

Why it is suitable for enterprises

Feature	Benefit for enterprises
Permissioned network	Participants are known and authorized, which meets compliance and KYC needs.
Privacy (channels, private data collections)	Only the parties involved see a transaction; competitors on the same network cannot.
Pluggable consensus (Raft, etc.)	No energy-hungry mining; fast, with immediate finality.
High performance and scalability	Execute-order-validate design gives high throughput and low latency.
Flexible endorsement policies	Organizations decide who must approve a transaction.
General-purpose chaincode	Written in standard languages, so developers need no special language.
No native cryptocurrency required	No gas fees; costs are predictable.
Modular architecture	Consensus, identity and storage can be swapped to fit the business.

Use cases: supply chain tracking, trade finance, healthcare records, banking and insurance.

Limitation: it is less decentralized than public chains such as Ethereum, because the network depends on trusted, known members.

Conclusion: Hyperledger Fabric suits enterprises because it combines blockchain's immutability and transparency with the privacy, access control, performance and compliance that businesses need.
